

SOFTWARE ENGINEERING LAB 3

Multi-Vendor Artisan E-Commerce Marketplace

Agile Project Planning and Sprint Simulation using Jira

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Problem Statement	Multi-Vendor Artisan E-Commerce Marketplace
Jira Project	Multi-Vendor Artisan E-Commerce Marketplace (key: MVAECM)
Jira Board	MVAECM board
Epics	4 (MVAECM-1 to MVAECM-4)
User Stories	12 (MVAECM-5 to MVAECM-16)
Sprints	MVAECM Sprint 1 and MVAECM Sprint 2 (one week each)

All Epics, User Stories, story points, priorities, sprint activity and reports shown in this document are taken from the student's own Jira project.

A. Epic and User Story Table

The table below lists all 4 Epics created in the Jira project and every User Story belonging to each Epic, together with its Jira issue key, story points, priority and the sprint in which it was worked on. All values are taken directly from the Jira evidence reproduced later in this document.

Epic	Epic Key	User Story	Issue Key	Story Points	Priority	Sprint
Vendor Storefront & Product Catalog	MVAECM-1	Create artisan storefront	MVAECM-5	3	High	Sprint 2
		Manage product catalog	MVAECM-6	5	High	Sprint 2
		Display product images	MVAECM-7	3	Medium	Backlog
Product Discovery & Shopping	MVAECM-2	Browse artisan products	MVAECM-8	3	High	Sprint 2
		View product details	MVAECM-9	3	High	Sprint 2
		Add products to cart	MVAECM-10	5	High	Sprint 1
Multi-Vendor Checkout & Payments	MVAECM-3	Place customer order	MVAECM-11	5	Highest	Sprint 1
		Process checkout payment	MVAECM-12	5	Highest	Sprint 1
		Track order status	MVAECM-13	3	Medium	Backlog
Order & Vendor Management	MVAECM-4	Split payment among vendors	MVAECM-14	8	Highest	Sprint 1
		Deduct platform commission	MVAECM-15	5	High	Backlog
		View vendor payout	MVAECM-16	3	Medium	Backlog
Total		12 User Stories across 4 Epics		51		

Source of the values above: Epic list - Figure 1; user stories, story points and priorities - Figures 2 and 3; sprint membership - Figures 5, 9, 10 and 13.

B. Evaluation Criteria and Where the Evidence Appears

Evaluation Criterion	Marks	Evidence in this document
Create Stories in Jira	4	Epic and User Story table (Part A); Section 1, Figures 1 and 2
Assign Story Points Planning Poker introduction	3	Section 2 - Planning Poker introduction and the agreed estimates; Story Points column of Figure 2 and of the table in Part A
Prioritize & Rank Backlog Set priority for each story; reorder backlog	2	Section 3, Figure 3 (priorities set and backlog reordered by rank)
Run a Mock Sprint Create Sprint 1; simulate progress; add comments	3	Section 4, Figures 4 and 5 (sprint created and started); Section 5, Figures 6 to 13 (progress simulation and comments); Section 6, Figures 14 to 17 (sprint summary)
Generate Burndown Chart	1	Section 7, Figures 18 and 19
Group Reflection	2	Section 8

1. Epics and User Stories

Four Epics (MVAECM-1 to MVAECM-4) were created for the Multi-Vendor Artisan E-Commerce Marketplace project, and twelve User Stories (MVAECM-5 to MVAECM-16) were created and linked to them, three stories per Epic.

The screenshot displays the Jira interface for the 'Multi-Vendor Artisan E-Commerce Marketplace' project. The 'MVAECM board' is shown in 'List' view. The board contains four epic items, each with a checkbox, a link icon, an ID, a title, an assignee, a reporter, a priority, and a status. All items are assigned to 'RACHIT MANE' and have a 'Medium' priority and 'To Do' status.

	Work	Assignee	Reporter	Priority	Status
☐	> MVAECM-1 Vendor Storefront & Product Catalog	Unassigned	RACHIT MANE	Medium	To Do
☐	> MVAECM-2 Product Discovery & Shopping	Unassigned	RACHIT MANE	Medium	To Do
☐	> MVAECM-3 Multi-Vendor Checkout & Payments	Unassigned	RACHIT MANE	Medium	To Do
☐	> MVAECM-4 Order & Vendor Management	Unassigned	RACHIT MANE	Medium	To Do

A notification in the bottom left corner states: 'Created Story "MVAECM-16"'. Below it, a link says 'View Story · Copy link'.

Figure 1 - The four Epics created in the Jira project: MVAECM-1 Vendor Storefront & Product Catalog, MVAECM-2 Product Discovery & Shopping, MVAECM-3 Multi-Vendor Checkout & Payments and MVAECM-4 Order & Vendor Management.

The screenshot displays the Jira interface for the 'Multi-Vendor Artisan E-Commerce Marketplace' project. The main backlog view shows twelve user stories, each with a unique ID, title, status, priority, and assignee. The stories are color-coded by epic: blue for MVAECM-5 to MVAECM-10, purple for MVAECM-11 to MVAECM-13, and orange for MVAECM-14 to MVAECM-16. The right-hand panel provides a detailed view of the selected story, MVAECM-7, 'Display product images', including its description, acceptance criteria, and attachments.

ID	Title	Status	Priority	Assignee
MVAECM-5	Create artisan storefront	To Do	3	RM
MVAECM-6	Manage product catalog	To Do	5	RM
MVAECM-7	Display product images	To Do	3	RM
MVAECM-8	Browse artisan products	To Do	3	RM
MVAECM-9	View product details	To Do	3	RM
MVAECM-10	Add products to cart	To Do	5	RM
MVAECM-11	Place customer order	To Do	5	RM
MVAECM-12	Process checkout payment	To Do	5	RM
MVAECM-13	Track order status	To Do	3	RM
MVAECM-14	Split payment among vendors	To Do	8	RM
MVAECM-15	Deduct platform commission	To Do	5	RM
MVAECM-16	View vendor payout	To Do	3	RM

Detail Panel for MVAECM-7: Display product images

Description: As an artisan vendor, I want to upload high-resolution product images so that shoppers can clearly view my products.

Acceptance Criteria:

- The vendor can upload product images.
- Images are displayed with the corresponding product.
- Shoppers can view the product images.

Attachments: Add attachment

Figure 2 - All twelve User Stories (MVAECM-5 to MVAECM-16) created in the backlog, each carrying its Epic colour tag. The detail panel shows story MVAECM-7 "Display product images" under Epic MVAECM-1, with its user-story description and acceptance criteria.

2. Story Points and Planning Poker Estimation

Planning Poker.

Story points for this project were estimated using the Planning Poker technique. In Planning Poker every team member privately selects an estimate for a user story from a Fibonacci-style deck (1, 2, 3, 5, 8, 13, ...); all estimates are revealed at the same time so that nobody anchors on anyone else's number, the highest and lowest estimates are discussed, and the round is repeated until the team converges on a single agreed value. The agreed value is then recorded against the story in Jira.

The values agreed for this project are 3, 5 and 8 - all Fibonacci values from the Planning Poker deck - and every one of the twelve user stories carries an estimate, giving a total of 51 story points for the product backlog. The agreed estimates are visible in the Story Points column of the Jira backlog in Figures 2 and 3, are listed per story in the table in Part A, and are summarised below.

Estimates agreed for the product backlog

Story Points	User Stories with this estimate	Number of Stories	Total Points
3	MVAECM-5, MVAECM-7, MVAECM-8, MVAECM-9, MVAECM-13, MVAECM-16	6	18
5	MVAECM-6, MVAECM-10, MVAECM-11, MVAECM-12, MVAECM-15	5	25
8	MVAECM-14	1	8
Total	12 User Stories	12	51

3. Priority and Backlog Ranking

A priority was set for every user story in Jira (Highest, High or Medium), and the backlog was then reordered so that the highest-priority and highest-value stories sit at the top and are pulled into the sprint first.

Backlog order before ranking (Figure 2): MVAECM-5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 - the order in which the stories were created.

Backlog order after ranking (Figure 3): MVAECM-14, 11, 12, 10, 6, 5, 8, 9, 15, 7, 13, 16 - the three Highest-priority stories (MVAECM-14, 11, 12) have been moved to the top of the backlog, followed by the High-priority stories, with the Medium-priority stories ranked last.

The screenshot shows the Jira backlog for the 'Multi-Vendor Artisan E-Commerce Marketplace' project. The backlog is sorted by priority and value. The top three items are MVAECM-14, MVAECM-11, and MVAECM-12, all with a priority of 'Highest' (indicated by a double chevron icon). The bottom three items are MVAECM-5, MVAECM-8, and MVAECM-9, all with a priority of 'Medium' (indicated by an equals sign icon). The right sidebar shows the details for the selected item, MVAECM-12, titled 'Process checkout payment'.

Item	Priority	Value
MVAECM-14	Highest	8
MVAECM-11	Highest	5
MVAECM-12	Highest	5
MVAECM-10	High	5
MVAECM-6	High	5
MVAECM-5	Medium	3
MVAECM-8	Medium	3
MVAECM-9	Medium	3
MVAECM-15	High	5
MVAECM-7	High	3
MVAECM-13	High	3
MVAECM-16	High	3

Figure 3 - The reordered and ranked product backlog. The priority icon on each row shows the priority set for that story in Jira (double chevron = Highest, single chevron = High, equals sign = Medium).

4. Sprint 1

MVAECM Sprint 1 was created from the top of the ranked backlog and started with a duration of one week. Four user stories worth 23 story points were committed to the sprint: MVAECM-14 (8), MVAECM-11 (5), MVAECM-12 (5) and MVAECM-10 (5).

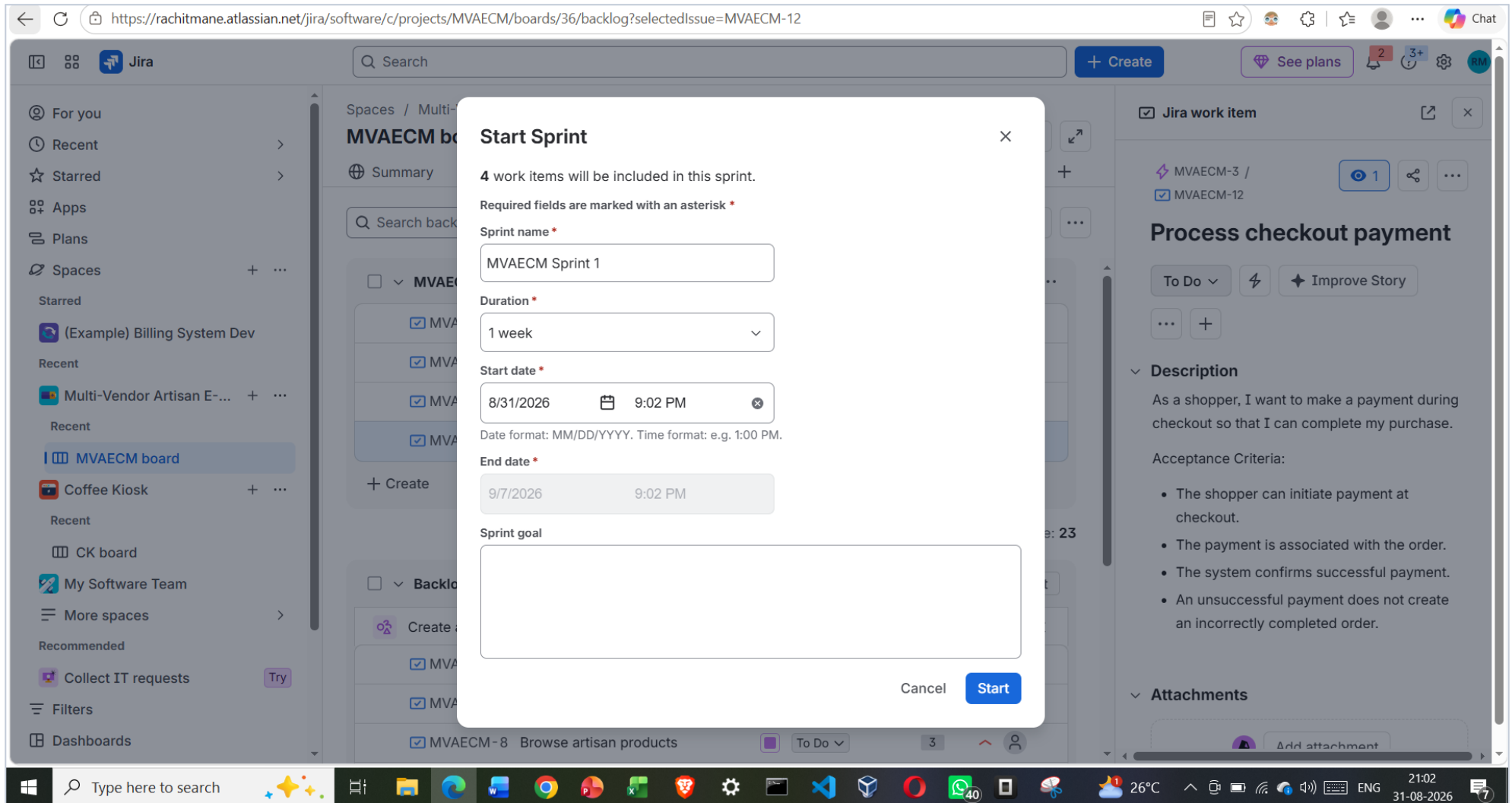


Figure 4 - Creating and starting MVAECM Sprint 1: sprint name "MVAECM Sprint 1", duration 1 week, start date 31/08/2026 9:02 PM, end date 07/09/2026 9:02 PM, with 4 work items included in the sprint.

The screenshot displays the Jira interface for the 'MVAECM board'. The left sidebar shows the 'For you' section with 'Recent' items including 'Multi-Vendor Artisan E-Commerce Marketplace' and 'Coffee Kiosk'. The main content area shows the 'MVAECM board' with tabs for 'Summary', 'List', 'Active sprints', 'Backlog', 'Development', 'Forms', 'Timeline', 'Docs', and 'Reports'. The 'Active sprints' tab is selected, showing a Kanban board with four columns: 'TO DO', 'IN PROGRESS', 'IN REVIEW', and 'DONE'. The 'TO DO' column contains four stories:

- Split payment among vendors** (MVAECM-14) with 8 points, assigned to RM.
- Place customer order** (MVAECM-11) with 5 points, assigned to RM.
- Process checkout payment** (MVAECM-12) with 5 points, assigned to RM.
- Add products to cart** (MVAECM-10) with 5 points, assigned to RM.

The bottom of the screen shows the Windows taskbar with various application icons and the system clock indicating 21:04 on 31-08-2026.

Figure 5 - MVAECM Sprint 1 started. The active sprint board shows the four committed stories in the TO DO column: MVAECM-14 Split payment among vendors (8), MVAECM-11 Place customer order (5), MVAECM-12 Process checkout payment (5) and MVAECM-10 Add products to cart (5).

5. Sprint Progress and Comments

Progress was simulated by moving the sprint stories across the board columns (To Do -> In Progress -> In Review -> Done) and by adding comments to the work items as the work progressed.

5.1 Sprint 1 progress and comments

The screenshot shows a Jira board for the project "Multi-Vendor Artisan E-Commerce Marketplace". The board is in "Active sprints" view. The columns are "TO DO", "IN PROGRESS", "IN REVIEW", and "DONE".

- TO DO (2 items):**
 - Split payment among vendors (MVAECM-14, 8 points)
 - Add products to cart (MVAECM-10, 5 points)
- IN PROGRESS (2 items):**
 - Place customer order (MVAECM-11, 5 points)
 - Process checkout payment (MVAECM-12, 5 points)
- IN REVIEW (0 items)**
- DONE (0 items)**

On the right, a "Jira work item" panel shows details for MVAECM-11. It includes a comment by RACHIT MANE: "Started implementation of the customer ordering workflow and validating the order creation process." The comment was made 1 second ago.

Figure 6 - Sprint 1 in progress: MVAECM-11 and MVAECM-12 moved to IN PROGRESS. Comment added by RACHIT MANE on MVAECM-11: "Started implementation of the customer ordering workflow and validating the order creation process."

← ↻ https://rachitmane.atlassian.net/jira/software/c/projects/MVAECM/boards/36?selectedIssue=MVAECM-12&sprintStarted=true

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MVAECM board Summary List Active sprints Backlog Development Forms More 3 +

Search board RM Version More Complete sprint Group: Story

TO DO 2 IN PROGRESS 2 IN REVIEW DONE

Split payment among vendors
Order & Vendor Ma...
8 MVAECM-14

Add products to cart
Product Discovery ...
5 MVAECM-10

Place customer order
Multi-Vendor Chec...
5 MVAECM-11

Process checkout payment
Multi-Vendor Chec...
5 MVAECM-12

+ Create

Jira work item

More fields Original estimate, Time tracki...

Automation Rule executions

Created 19 minutes ago Updated 1 minute ago Configure

Activity

Comments More

Add a comment...
Suggest a reply... Who is w >

Pro tip: press M to comment

RACHIT MANE
1 second ago

Working on checkout payment processing and validating successful and unsuccessful payment scenarios.

← 👍 😊 📎 ...

Type here to search 26°C 21:07 31-08-2026

Figure 7 - Comment added by RACHIT MANE on MVAECM-12: "Working on checkout payment processing and validating successful and unsuccessful payment scenarios."

← ↻ 🔒 https://rachitmane.atlassian.net/jira/software/c/projects/MVAECM/boards/36?sprintStarted=true

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Summary List **Active sprints** Backlog Development Forms Timeline Docs Reports +

Search board RM Version Epic Quick filters Complete sprint Group: Story

TO DO 2

Split payment among vendors

Order & Vendor Management

MVAECM-14 8

Add products to cart

Product Discovery & Shopping

MVAECM-10 5

+ Create

IN PROGRESS

IN REVIEW 1

Place customer order

Multi-Vendor Checkout & Payments

MVAECM-11 5

DONE 1 ✓

Process checkout payment

Multi-Vendor Checkout & Payments

MVAECM-12 5

+ 21:08 31-08-2026

Figure 8 - Sprint 1 progress: MVAECM-11 Place customer order moved to IN REVIEW and MVAECM-12 Process checkout payment moved to DONE.

← ↻ https://rachitmane.atlassian.net/jira/software/c/projects/MVAECM/boards/36?sprintStarted=true

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MVAECM board ...

Summary List **Active sprints** Backlog Development Forms Timeline Docs Reports +

Search board RM Version Epic Quick filters Complete sprint Group: Story

TO DO IN PROGRESS IN REVIEW DONE 4 ✓

+ Create

Process checkout payment
Multi-Vendor Checkout & Payments
MVAECM-12 5

Place customer order
Multi-Vendor Checkout & Payments
MVAECM-11 5

Split payment among vendors
Order & Vendor Management
MVAECM-14 8

Add products to cart
Product Discovery & Shopping
MVAECM-10 5

Windows Taskbar: Type here to search, 26°C, 21:09, 31-08-2026

Figure 9 - End of Sprint 1: all four committed stories (MVAECM-12, MVAECM-11, MVAECM-14 and MVAECM-10) completed in the DONE column.

5.2 Sprint 2 progress and comments

After Sprint 1 was completed, the next four stories from the ranked backlog were taken into MVAECM Sprint 2 (MVAECM-6, MVAECM-5, MVAECM-8 and MVAECM-9, worth 14 story points) and the same progress simulation and commenting was carried out. MVAECM Sprint 2 runs over the same one-week window as Sprint 1, as shown by the sprint burndown in Figure 19.

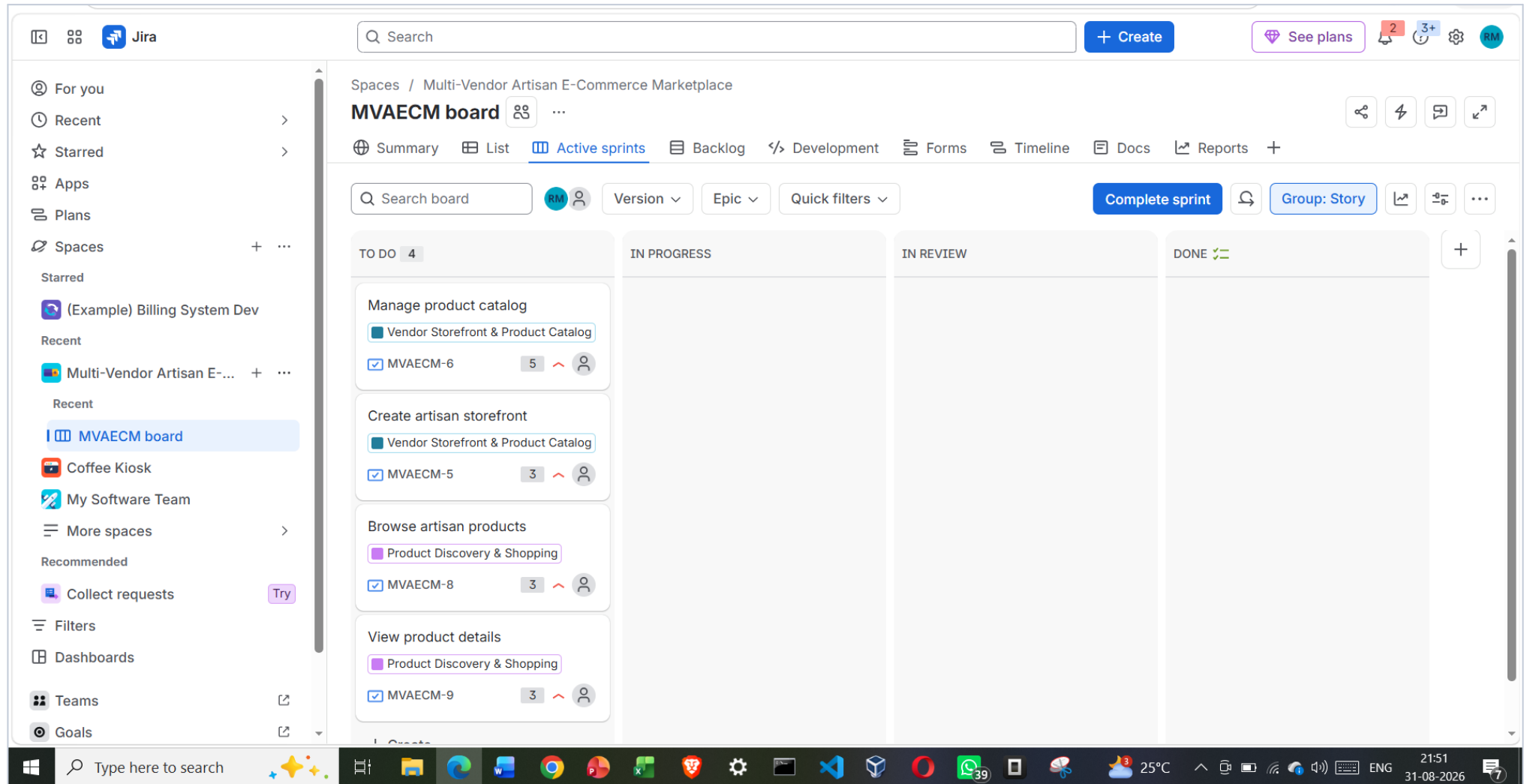


Figure 10 - MVAECM Sprint 2 board at the start of the sprint, with MVAECM-6 Manage product catalog (5), MVAECM-5 Create artisan storefront (3), MVAECM-8 Browse artisan products (3) and MVAECM-9 View product details (3) in the TO DO column.

The screenshot shows a Jira board for the 'Multi-Vendor Artisan E-Commerce Marketplace' project. The board is titled 'MVAECM board' and is in the 'Active sprints' view. The board has four columns: 'TO DO' (3 items), 'IN PROGRESS' (1 item), 'IN REVIEW' (0 items), and 'DONE' (0 items). The 'IN PROGRESS' column contains a card for 'Manage product catalog' with a sub-card 'Vendor Storefront ...' and a checkbox for 'MVAECM-6'. The 'TO DO' column contains cards for 'Create artisan storefront' (with sub-card 'Vendor Storefront ...' and checkbox 'MVAECM-5'), 'Browse artisan products' (with sub-card 'Product Discovery...' and checkbox 'MVAECM-8'), and 'View product details' (with sub-card 'Product Discovery...'). The right sidebar shows the 'Jira work item' details for 'MVAECM-6', including a comment by 'RACHIT MANE' (RM) stating: 'Started implementation of the product catalog functionality. Working on vendor product management and catalog organization.' The bottom of the screen shows a Windows taskbar with various application icons and system information.

Figure 11 - Sprint 2 in progress: MVAECM-6 moved to IN PROGRESS. Comment added by RACHIT MANE: "Started implementation of the product catalog functionality. Working on vendor product management and catalog organization."

The screenshot displays the Jira interface for the 'MVAECM board'. The board is organized into four columns: 'TO DO', 'IN PROGRESS', 'IN REVIEW', and 'DONE'. The 'IN PROGRESS' column contains two items: 'Manage product catalog' (MVAECM-6) and 'Create artisan storefront' (MVAECM-5). A comment by RACHIT MANE is visible on the right side of the board.

Board Columns and Items:

- TO DO (2 items):**
 - Browse artisan products (Product Discovery ...)
 - View product details (Product Discovery ...)
- IN PROGRESS (2 items):**
 - Manage product catalog (Vendor Storefront ...)
 - Create artisan storefront (Vendor Storefront ...)
- IN REVIEW**
- DONE (2 items)**

Comment by RACHIT MANE:

Storefront development is in progress. Initial vendor storefront structure and required functionality are being implemented.

Figure 12 - Sprint 2 in progress: MVAECM-5 also moved to IN PROGRESS. Comment added by RACHIT MANE: "Storefront development is in progress. Initial vendor storefront structure and required functionality are being implemented."

← ↻ 🔒 https://rachitmane.atlassian.net/jira/software/c/projects/MVAECM/boards/36?sprintStarted=true

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Search board RM Version Epic Quick filters Complete sprint Group: Story

TO DO + Create

IN PROGRESS

IN REVIEW

DONE 4

Manage product catalog

- Vendor Storefront & Product Catalog
- MVAECM-6 5

Create artisan storefront

- Vendor Storefront & Product Catalog
- MVAECM-5 3

Browse artisan products

- Product Discovery & Shopping
- MVAECM-8 3

View product details

- Product Discovery & Shopping
- MVAECM-9 3

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Figure 13 - End of Sprint 2: all four sprint stories (MVAECM-6, MVAECM-5, MVAECM-8 and MVAECM-9) completed in the DONE column.

The Jira Sprint Report for MVAECM Sprint 1 shows the sprint burndown together with the status report of completed issues. Sprint 1 was completed with all 23 committed story points delivered and no incomplete issues.



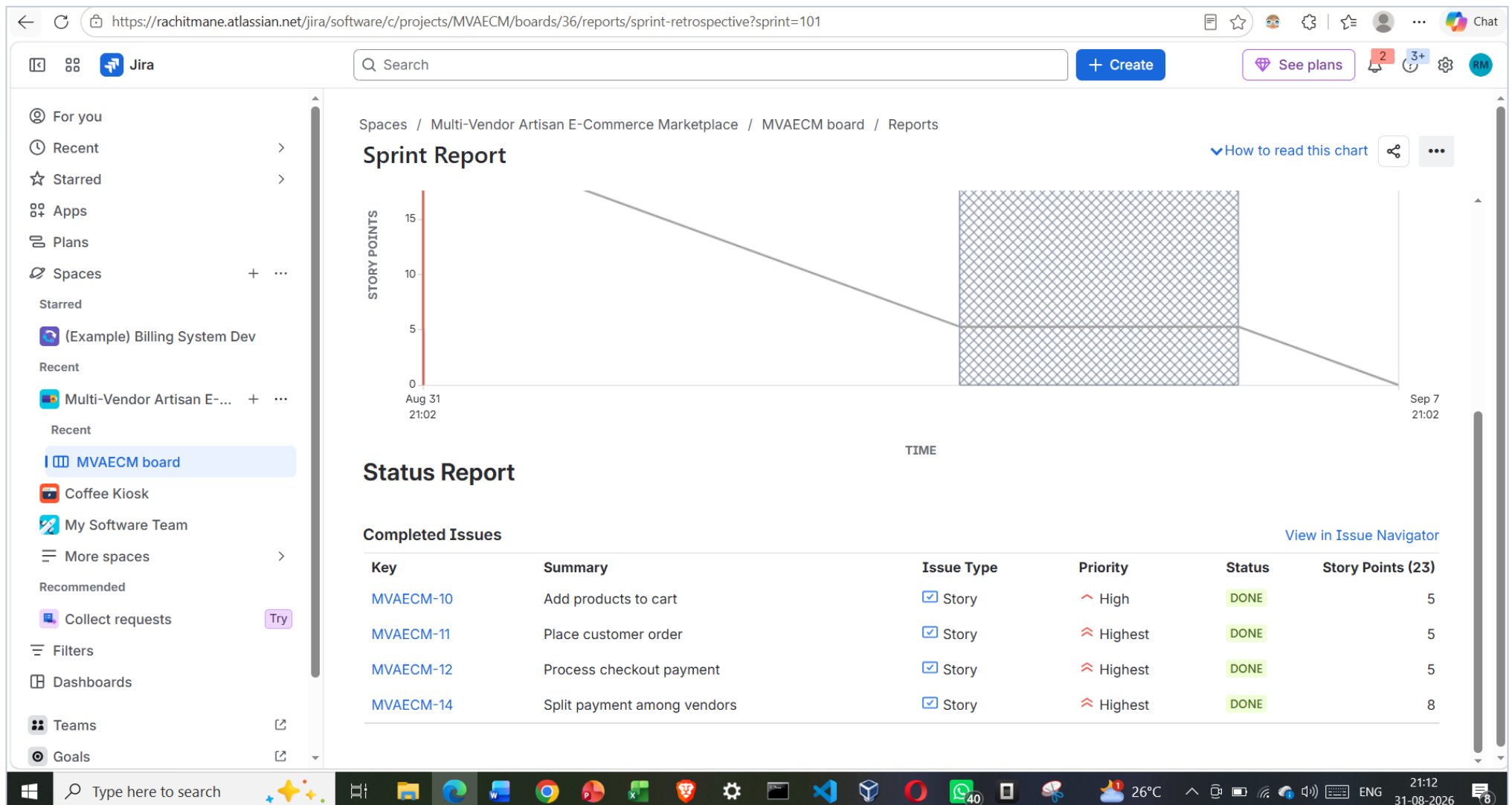


Figure 15 - Sprint Report status report - completed issues: MVAECM-10 Add products to cart (High, 5), MVAECM-11 Place customer order (Highest, 5), MVAECM-12 Process checkout payment (Highest, 5) and MVAECM-14 Split payment among vendors (Highest, 8). Total 23 story points.

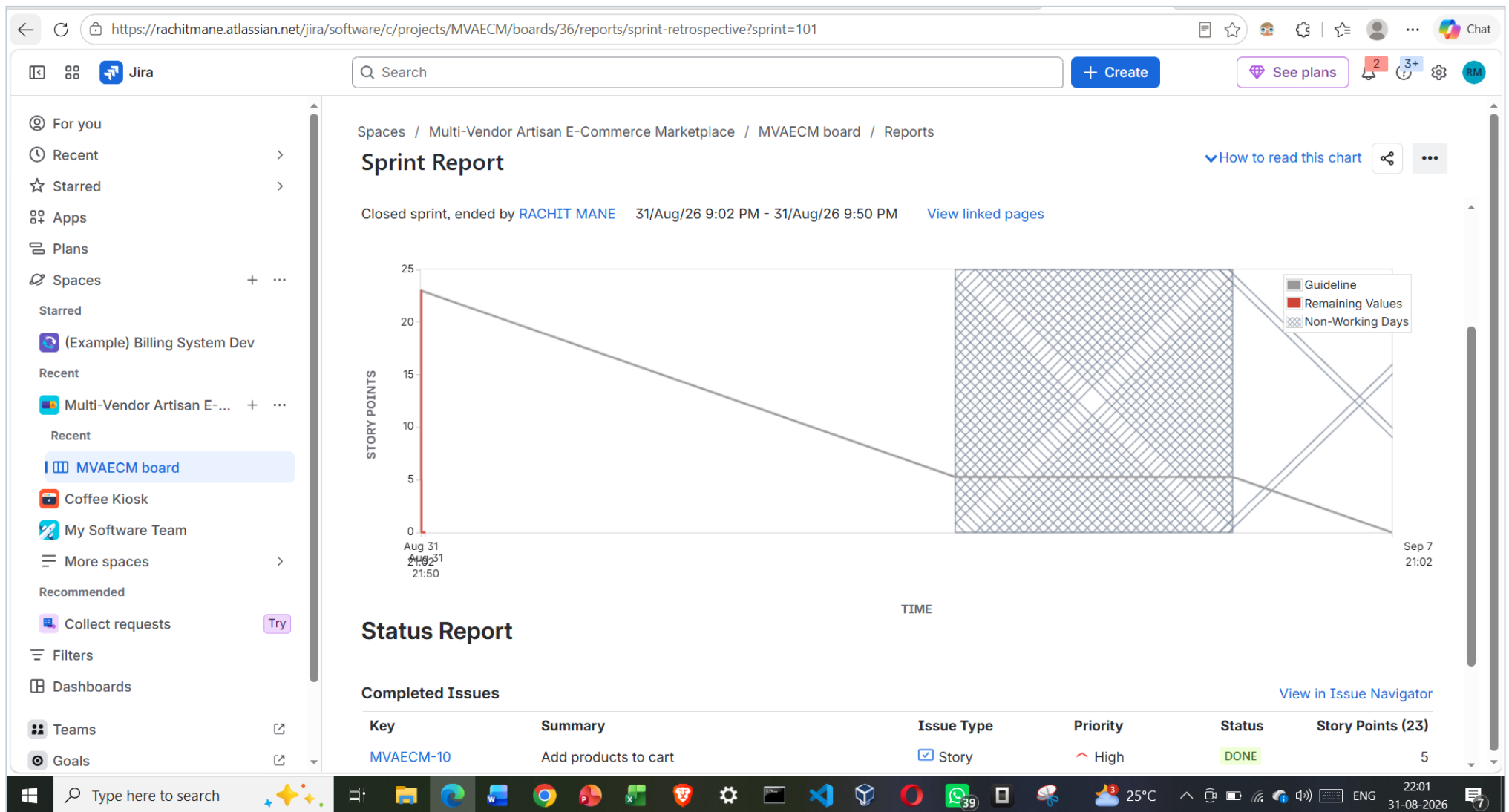


Figure 16 - Sprint 1 completed: "Closed sprint, ended by RACHIT MANE - 31/Aug/26 9:02 PM - 31/Aug/26 9:50 PM", with the completed sprint burndown.

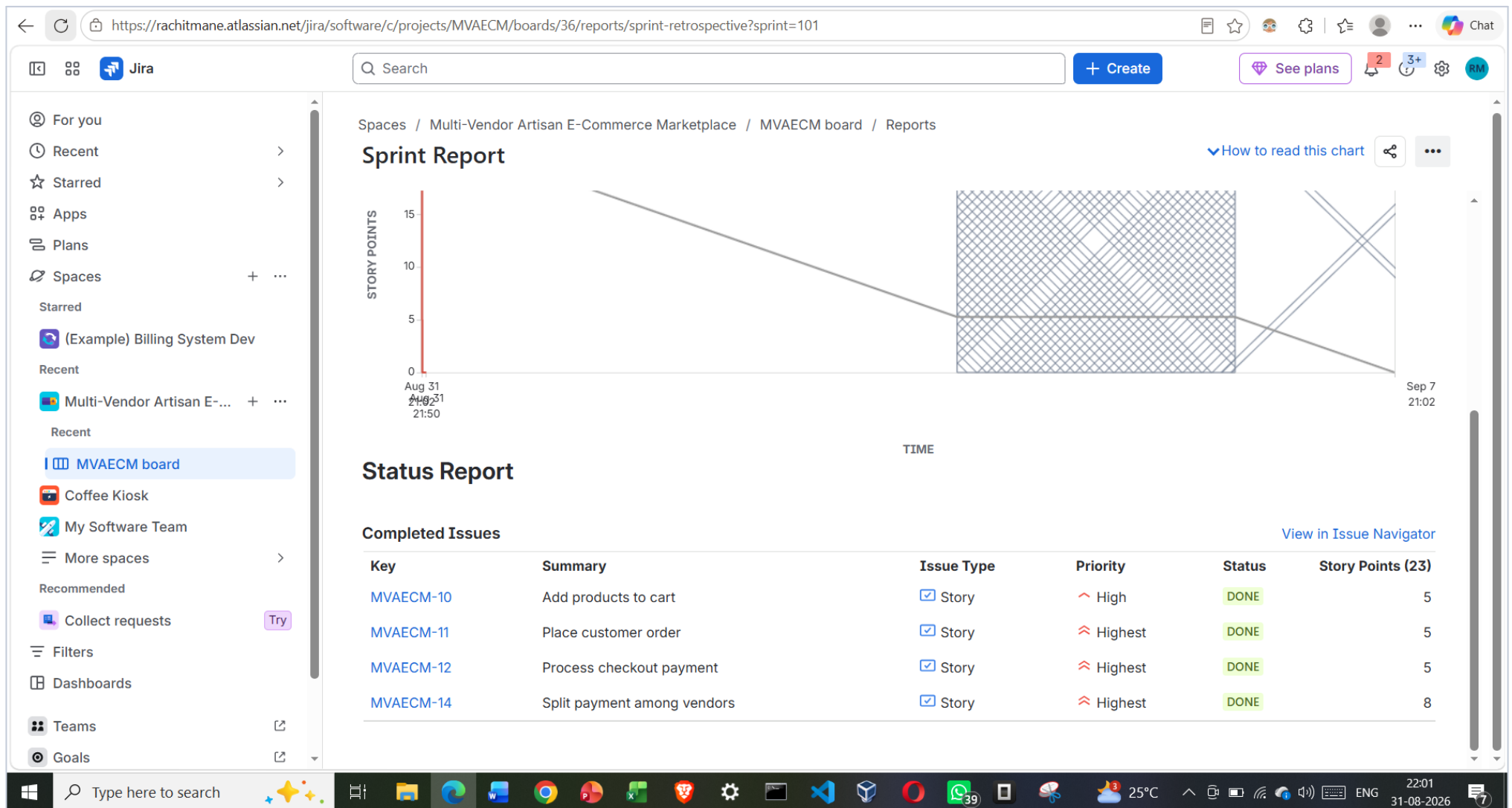


Figure 17 - Sprint 1 completion summary: all four committed stories are listed under Completed Issues with status DONE and a total of 23 story points delivered.

7. Burndown Chart

The burndown chart for each sprint was generated from Jira Reports and plots the remaining story points against time, with the ideal guideline, the remaining values and the non-working days shown.

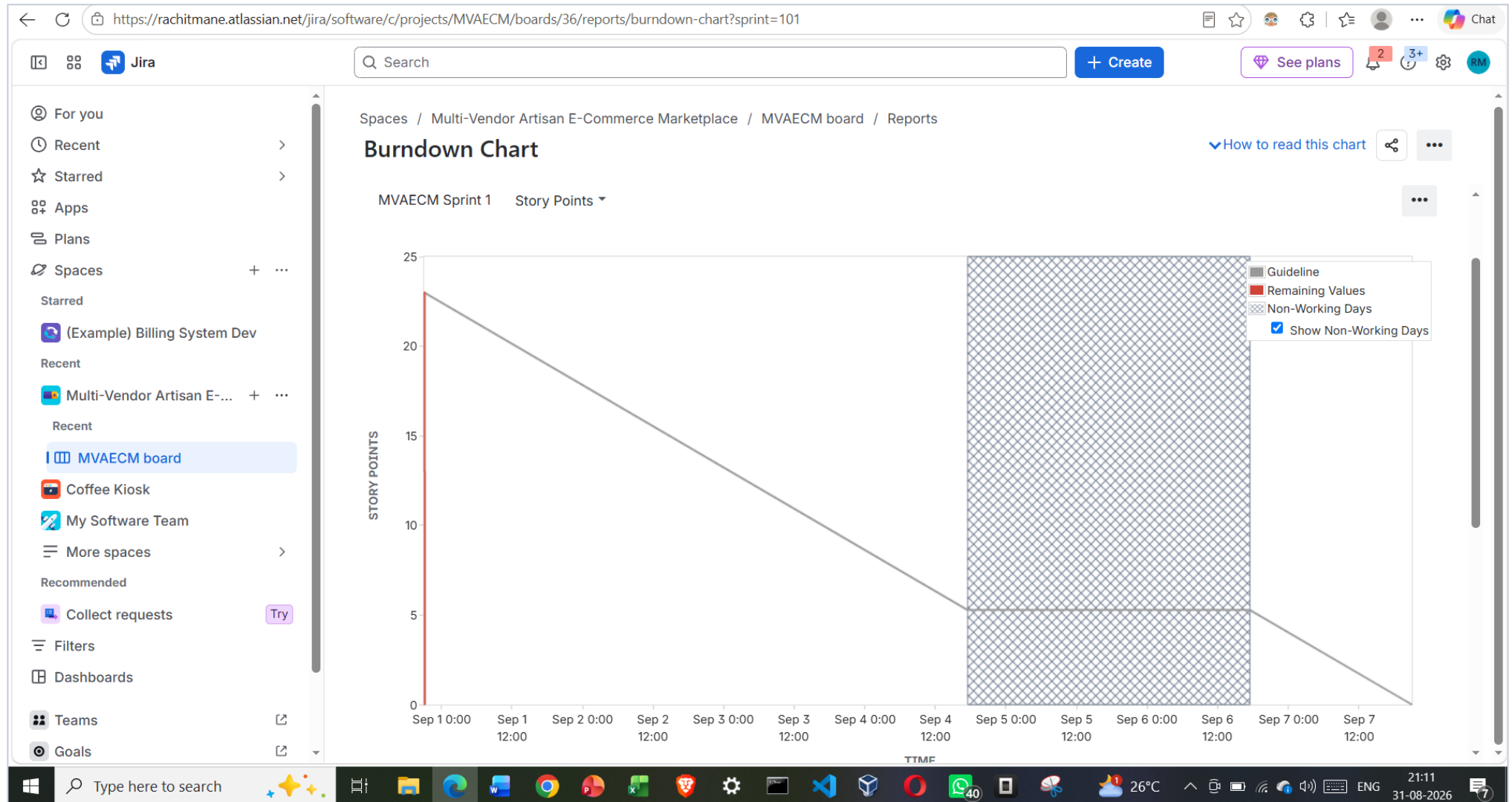


Figure 18 - Burndown chart for MVAECM Sprint 1, measured in story points, starting from the 23 story points committed to the sprint.

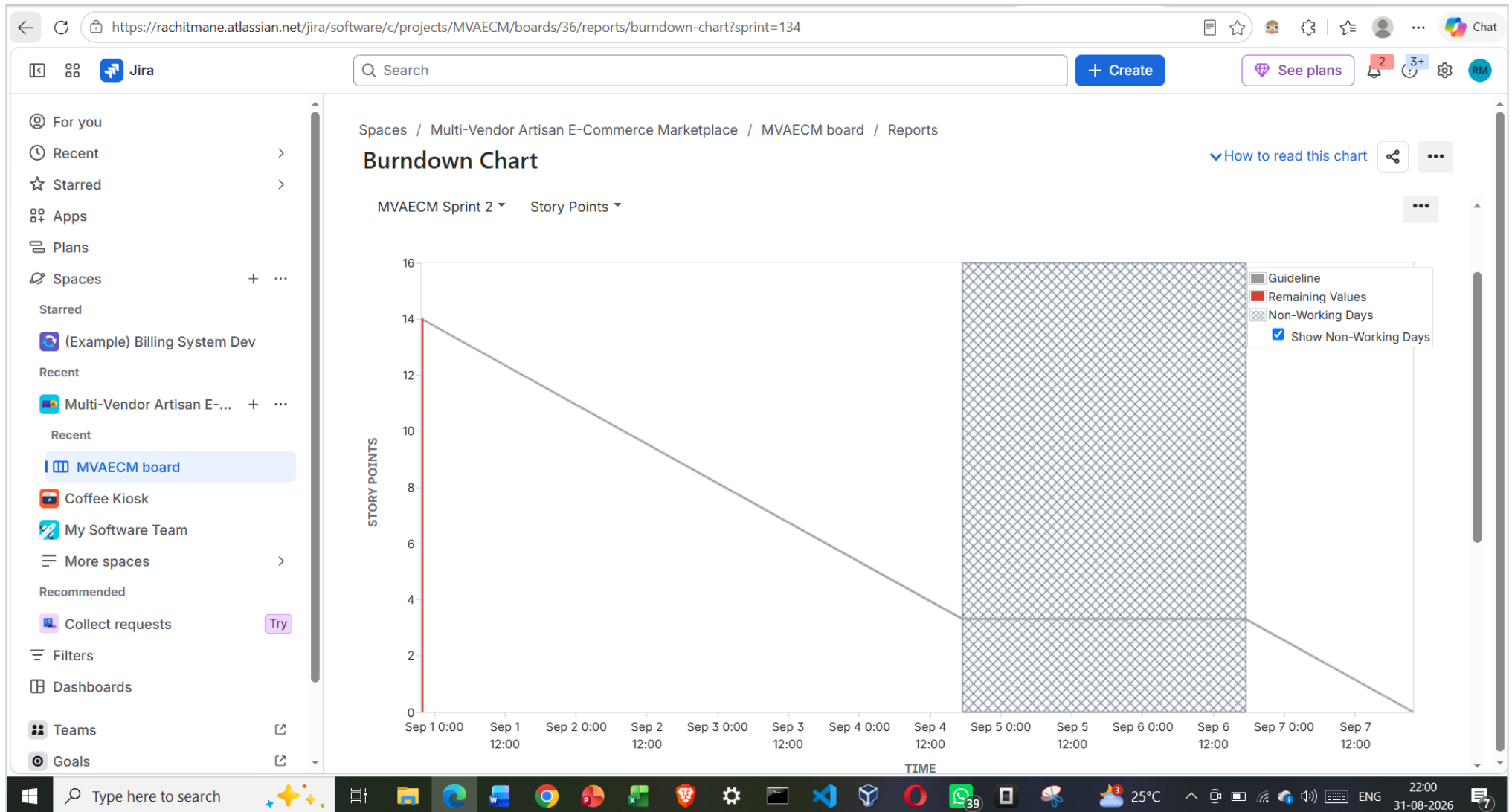


Figure 19 - Burndown chart for MVAECM Sprint 2, measured in story points, starting from the 14 story points committed to the sprint.

8. Group Reflection

What we learned from using Jira

Modelling the Multi-Vendor Artisan E-Commerce Marketplace as four Epics and twelve User Stories showed us how a product backlog is actually structured: every story is linked to the Epic it belongs to, so the board always makes clear which part of the product a piece of work supports. Writing each item as a user story with explicit acceptance criteria forced us to agree what "done" meant before any work was started, rather than after.

Planning Poker and story-point estimation

Estimating with Planning Poker moved us away from guessing hours and towards comparing stories against each other. Revealing the estimates at the same time stopped the first number spoken from anchoring the rest of the group, and the discussion that followed a disagreement was usually more useful than the number we finally agreed on. The Fibonacci values we settled on - 3, 5 and 8 - gave us 51 story points across the backlog, which is what made a realistic sprint commitment possible.

Backlog prioritisation and ranking

Setting a priority of Highest, High or Medium on every story and then reordering the backlog by rank made sprint planning almost mechanical, because the stories at the top were simply the ones we pulled in first. The clearest lesson was that priority and rank are not the same thing: MVAECM-14, MVAECM-11 and MVAECM-12 all carry the priority Highest, but the backlog still has to place them in one definite order.

Sprint planning and progress tracking

Sprint 1 took the top four ranked stories (23 story points) and Sprint 2 the next four (14 story points), each over a one-week sprint. Moving cards through To Do, In Progress, In Review and Done, and adding a comment on a work item when we picked it up, left a visible history on the item itself instead of a status that only existed in conversation, and the burndown chart then reported that progress for us automatically.

What went well

Every story committed to both sprints reached Done - 23 story points in Sprint 1 and 14 in Sprint 2, with no incomplete issues in the Sprint Report. Grouping the scenario into four Epics kept related stories together and made the backlog easy to read, and because the backlog was ranked before the sprint started we never had to argue about what to work on next.

What was challenging

Agreeing an estimate for MVAECM-14 Split payment among vendors was the hardest part of estimation: at 8 story points it is the largest item in the backlog and it covers both the commission deduction and the payout calculation, which suggests it could have been split into smaller stories. Choosing an order among several stories that all carried the priority Highest also took discussion. Because the sprint was simulated within a single working session, the burndown drops sharply near the end instead of falling steadily, so it reflects the outcome of the sprint better than the pace of it.

What we would improve in the next sprint

We would split any story estimated above 5 points into smaller stories so that progress becomes visible earlier in the sprint; assign an owner to each story rather than leaving work items unassigned; move cards across the board as the work actually happens so that the burndown tracks the guideline more closely; and record acceptance criteria on every story, as we did for MVAECM-7 and MVAECM-12, so that each story can be verified independently.